

Facial Expression Spotting Based on Optical Flow Features

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ABSTRACT

The purpose of micro expression (ME) and macro expression (MaE) spotting task is to locate the onset and offset frames of MaE and ME clips. Compared with MaEs, MEs are shorter in duration and lower in intensity, which makes MEs harder to be spotted. In this paper, we propose an efficient pipeline based on optical flow features to spot MEs and MaEs. We crop and align the faces and select the eyebrows area, nose area, and mouth area as our regions of interest to exclude the interference of extraneous factors on the face expression representation. Then, we extract the optical flow in these regions and enhance the features of the expressions in the optical flow with low-pass filter and EMD method. Finally, the sliding window method is used to locate the peaks of optical flow features and get the intervals containing MEs or MaEs. We evaluate the performance of our method on the MEGC2022-TestSet including 10 long videos from SAMM and CAS(ME)³ and achieve the first place in the MEGC2022 Challenge. The results prove the effectiveness of our method.

CCS CONCEPTS

• Computing methodologies → Computer vision tasks.

KEYWORDS

Optical Flow, Expression Spotting, Micro expression, Macro expression

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1 INTRODUCTION

Facial expression is one of the important manifestations of our human beings to convey our inner emotions and psychological changes. It contains a lot of information even more than body movements and language expressions, so it can be used to analyze a person's mental activity and emotional state. Facial expressions can be divided into two categories: macro expression (MaE) and micro expression (ME). Although the boundary between them is difficult to distinguish, researchers generally believe that: MaE, also known as the regular facial expression, lasts from 0.5s to 4.0s[24] and has higher intensity, while ME occurs in less than 0.5s and has lower intensity[5]. It is challenging to spot the expression intervals in videos.

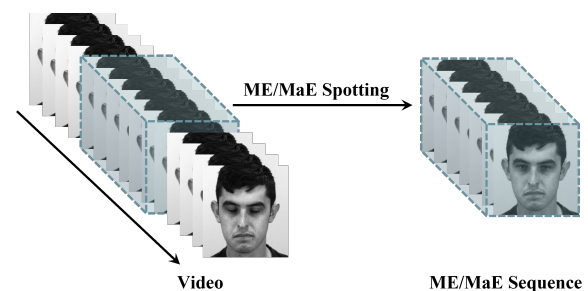


Figure 1: ME and MaE spotting process.

The purpose of ME and MaE spotting task is to locate the onset and offset frames of MaE and ME clips as shown in Figure 1. It is an important step before ME and MaE recognition that may seriously affect the accuracy and it is viewed as an important method to provide valuable data and can be applied in various scenarios, such as law enforcement, lie detection[17], political psychology[21] and medical care[7]. Datasets are vital for the research and there are many benchmark datasets for MaE and ME spotting: CAS(ME)²[19], SAMM Long Videos (SAMM-LV)[26], SMIC-E-long[22], SAMM[3] and CAS(ME)³[13]. Figure 2 shows an example in the SAMM-LV[26] dataset.

The optical flow method uses the temporal variation of pixels in the image sequence and the correlation between adjacent frames to find the correspondence between the previous frame and the current frame. It can be used to calculate subtle movements of the

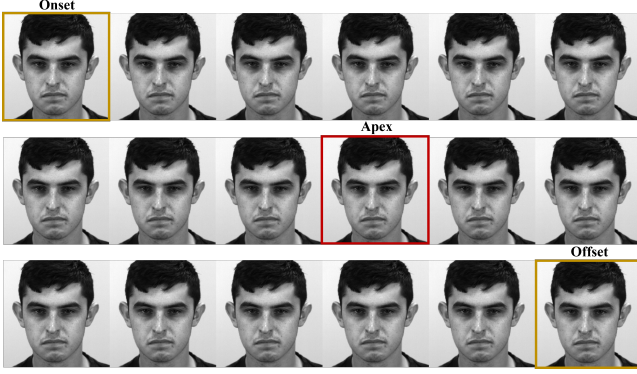


Figure 2: A sequence of image frames (ordered from left to right, top to bottom) of an expression from SAMM-LV[26], with the onset, apex and offset frame indications.

facial area. This paper proposes a ME and MaE spotting method based on the local optical flow features method. The framework of the proposed method is shown in Figure 3. We first crop and align the faces in the frames and select eyebrows area, nose area, and mouth area from the human face as our regions of interest. Then the sliding window method is utilized to locate the peaks of optical flow features and get intervals containing expressions. Finally, the intervals with greater Intersection over Union (IoU) would be fused to get the final expression intervals.

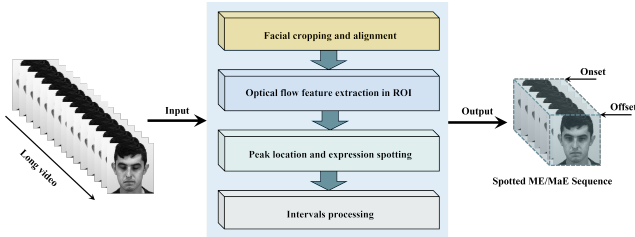


Figure 3: The framework of our proposed method.

2 RELATED WORK

2.1 Optical Flow

Optical flow estimation requires accurate per-pixel localization and finding the correspondence between two input images. Horn and Schunck[10] proposed the variational methods for the optical flow estimation. Many improvements have been introduced[1, 15], and combinatorial matching has been integrated into the variational approach[2, 23]. Machine learning techniques have been applied to optical flow before. Rosenbaum et al.[20] modeled local statistics of optical flow with Gaussian mixture models. Gunnar Farneback[8] proposed a two-frame motion estimation algorithm and the evaluation on Yosemite sequence showed good results.

As a micro-expression occurs, the movements of the human face are small with short duration. These changes are difficult to represent in RGB images. Therefore, using optical flow motion to

describe facial motion is very helpful for feature extraction of micro expressions.

2.2 MaE and ME Spotting Methods

As an important components of ME analysis, the ME and MaE spotting task is gaining attention. Moilanen[14] utilized Chi-Squared dissimilarity to calculate the distribution difference between the Local Binary Pattern (LBP) histogram of the current feature frame and the averaged feature frame. Duque et al.[4] proposed the Riesz pyramid-based method. The optical flow-based method proposed by [9] and [30]. Pan, et al.[18] proposed a local bilinear convolutional neural network model to classify an image from video into ME, MaE and natural.

In Facial Micro-Expression (FME) Workshop and Challenge 2021, He[29] proposed an optical flow features method to spot MEs and MaEs with a high precision and Recall. Yu et al.[28] used LSSNet and Yang et al.[25] utilized facial action units detection to construct an end-to-end deep learning framework for the MEs and MaEs spotting task. All of these methods exceeded the MEGC2021 baseline[27] which proved the effectiveness of these methods.

In our approach, there are two main core points. The first one is the selection of some local facial regions to extract the optical flow, which reduces the impact caused by the motion of other regions of the face. The second one is the use of low-pass filter and Empirical Mode Decomposition (EMD)[11] method to clean the optical flow features to make the expression spotting more accurate. The results prove the effectiveness of our proposed method.

3 METHODOLOGY

As shown in Figure 3, our pipeline can be divided into 4 parts: Facial cropping and alignment, Optical flow feature extraction in regions of interest, Peak location and expression spotting and Intervals processing. In this section, we will cover each part in detail.

3.1 Data Process

Face Clip. It is necessary to preprocess images in the dataset since facial expressions are easily disturbed by other factors. We use the Dlib toolkit[12] to detect the Facial Landmarks in each frames as a basis to calculate the size of each face and make a crop. The face images in the SAMM are resized to 320×320 and the faces in the CAS(ME)³ are resized to 256×256.

Cutting Box Adjustment. He[29] proposed that using cutting box adjustment had achieved a significant improvement in the F_1 score of ME and MaE spotting task using optical flow features. We also assume that the expression is not related to the movement of the nose tip area. $|\rho_{nose}|$ represents the amplitude of the optical flow vector ρ_{nose} of the nose tip area. When $|\rho_{nose}|$ exceeds 1 pixel, the cutting box moves by the distance of $|\rho_{nose}|$ in the opposite direction of the optical flow vector ρ_{nose} .

3.2 Optical Flow Extraction

Regions of Interest. According to the Facial Coding System[6], facial expressions can be broken down into individual components of muscle movement. We use the change of optical flow intensity in the regions of interest (ROI) to represent the movement of facial muscles. Different ROI contribute differently to expressions. As

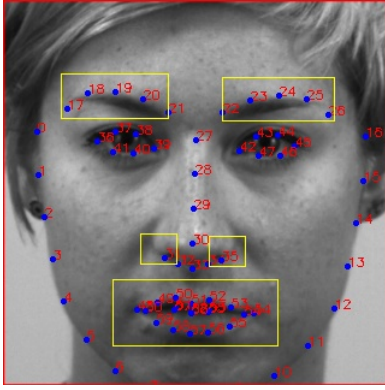


Figure 4: The cutting box and regions of interest.

shown in Figure 4, in order to minimize the maximum influence of distracting factors, we choose the eyebrows area, both sides of the nose area and the mouth area as our ROI. Intuitively, the motion of the face is holistic and can be represented by the five regions in Figure 4, and we have verified this experimentally.

Optical Flow Extraction. We extract the ROI from the reference frame and the subsequent frame, and calculate the dense optical flow of them using Gunnar Farneback algorithm[8] (optical flow is calculated for each pixel).

3.3 Expression Spotting

Slide window setting. Since the frame rate of SAMM[3] is 200 fps, the amplitude of the optical flow vectors of two adjacent frames is relatively small. So we conduct a sampling on SAMM: the first frame is selected from every 7 frames to make sure the frame rate of both SAMM and CAS(ME)³[13] is approximately equal to 30 fps. We choose a window size of 200 frames, i.e., we assume that an expression will not exceed 200 frames from the beginning to the end. Finally, for the 200 frames in the window, we calculate the optical flow between the current frame and the first frame, and subtract the optical flow of the nose region from the optical flow of each region as the final optical flow features.

Denoising. We use FFT[16] method to process the optical flow features, and set a low-pass filter to filter out the high-frequency noise in the optical flow signal to get the $flow_1$. Then we use the EMD method to decompose the optical flow signal $flow_1$ into individual Intrinsic Mode Functions (IMFs) and remove the high frequency Intrinsic Mode Function from the IMFs to reconstruct the optical flow signal $flow_2$. Thus, we get two relatively clean optical flow features. Figure 5 shows the comparison of the optical flow signal through the low-pass filter. We can find that the low-pass filter filters out the noise signal very well.

Spotting. We first set a threshold θ_1 , only when the intensity of the optical flow is greater than θ_1 can be selected as candidate expression intervals. Among these intervals, we also filter out the intervals where the difference between the maximum and minimum values of the optical flow intensity is greater than the threshold θ_2 . To make the localization more accurate, we set the threshold θ_3 . Only when the amplitude of the optical flow signal within these

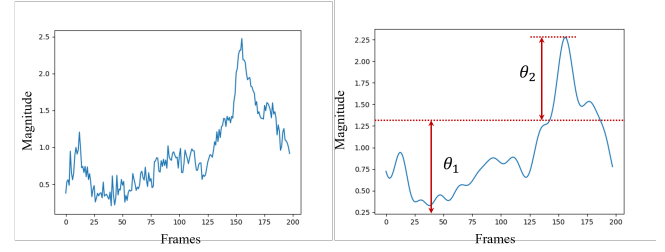


Figure 5: Comparison of optical flow characteristics signal through low-pass filter. The left figure shows the optical flow features before filtering, and the right figure shows the optical flow features after filtering.

intervals after reconstruction with IMFs is greater than θ_3 , we use these intervals as the final expression intervals.

Intervals fusion. Through the previous steps, we get many expression intervals and then calculate the IoU between these intervals. Finally, we fuse the intervals with IoU greater than a threshold to get the final expression intervals.

4 EXPERIMENTS

4.1 Dataset

CAS(ME)²[19] and SAMM-LV[26] are used to explore the general patterns of expression optical flow signals. We use these patterns for expression interval prediction in the testing set. In the CAS(ME)², there are 22 subjects and 98 long videos, and the average duration is 148s. The samples contain 300 MaEs and 57 MEs in total. In SAMM-LV, there are 147 videos which contain 343 MaEs and 159 MEs in total. Both of these two datasets provide the onset, apex and offset index for expressions in the excel file and the eye blinks are also labeled with onset and offset time.

Furthermore, the unseen testing set (MEGC2022-TestSet) is used to evaluate the performance of our method. It contains 10 long video, including 5 long videos from SAMM[3] (SAMM Challenge dataset) and 5 clips cropped from different videos in CAS(ME)³[13]. The frame rate for SAMM Challenge dataset is 200 fps and the frame rate for CAS(ME)³ is 30 fps.

4.2 Setup

We analyze the optical flow signal intensity of the training set and perform a parameter search to determine the approximate range of parameters, and then fine-tune the parameters on the test set to obtain the final parameters. We set the size of the sliding window to 200, and move it 80-120 frames at a time. For SAMM, θ_1 is set to 1.3, θ_2 is set to 1.4, θ_3 is set to 0.8. For CAS(ME)³, θ_1 is set to 1.6, θ_2 is set to 2.0, θ_3 is set to 0.5. Finally, the IoU threshold is set to 0.2, and whenever the IoU of two expression intervals exceeds 0.2, then the two intervals are merged. Experiments are also conducted using MindSpore.

Table 1: The leaderboard of ME and MaE spotting task.

Team	Both			SAMM			CAS(ME) ³		
	F1 Score	Precision Score	Recall Score	F1 Score	Precision Score	Recall Score	F1 Score	Precision Score	Recall Score
Ours	0.3514	0.3210	0.3881	0.3846	0.4762	0.3226	0.3333	0.2667	0.4444
ZhaoYuan	0.3467	0.3133	0.3881	0.3265	0.4444	0.2581	0.3564	0.2769	0.5000
whcold	0.2975	0.3333	0.2687	0.2264	0.2727	0.1935	0.3529	0.3750	0.3333
HIT_BINE	0.2879	0.2923	0.2836	0.2963	0.3478	0.2581	0.2821	0.2619	0.3056
genbing	0.2586	0.3061	0.2239	0.2381	0.4545	0.1613	0.2703	0.2632	0.2778
jiateng_liu	0.2500	0.2338	0.2687	0.2105	0.2308	0.1935	0.2759	0.2353	0.3333
AIPL_xxl	0.2290	0.2344	0.2239	0.1364	0.2308	0.0968	0.2759	0.2353	0.3333
xueyunlong	0.1613	0.1754	0.1493	0.2105	0.2308	0.1935	0.1194	0.1290	0.1111
Baseline [27]	0.1351	0.1235	0.1493	0.1176	0.0845	0.1935	0.1739	0.4000	0.1111
XSBANK	0.1294	0.1068	0.1642	0.1750	0.1429	0.2258	0.0889	0.0741	0.1111
xuke2021	0.0921	0.0824	0.1045	0.0556	0.0488	0.0645	0.1250	0.1136	0.1389

4.3 Metrics

We use the IoU threshold method to measure the relationship between a spotted interval and the ground-truth interval. Each ground-truth interval can only correspond to at most one spotted interval. When the relationship between the spotted interval and the ground-truth interval satisfies the following formula, we believe that the classification result belongs to true positive (TP) :

$$\frac{W_{Spotted} \cap W_{GroundTruth}}{W_{Spotted} \cup W_{GroundTruth}} \geq k \quad (1)$$

where k is a threshold, generally can be set to 0.5, $W_{GroundTruth}$ represents the ground truth of the MaE or ME interval (onset-offset), and $W_{Spotted}$ represents the spotted interval of the MaE or ME interval (onset-offset). If the condition is not fulfilled, the spotted interval is regarded as false positive (FP) . If the ground-truth interval missed, it is regarded as false positive (FN) .

Precision is the probability of a positive sample out of all the samples that are predicted to be positive. Recall means the probability of a positive sample being predicted out of an actual positive sample. For the expression spotting task, we of course want both precision and recall to be high, so we also use F_1 score to evaluate the performance of our method.

$$Precision = \frac{TP}{TP + FP} \quad (2)$$

$$Recall = \frac{TP}{TP + FN} \quad (3)$$

$$F_1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (4)$$

4.4 Results

Table 2 shows the effect of using denoising method by evaluating on CAS(ME)². From the experimental results, the method of low-pass filter to remove high-frequency noise can significantly improve the precision and recall score in expression spotting task. This indicates that the optical flow features with filtered noise can indeed better represent the generation of expressions. The use of EMD method to further assist in denoising works, but not very well.

We use the unseen testing set (MEGC2022-testSet) to evaluate the performance of our method. For CAS(ME)³, the F_1 score of the proposed method is 0.3333. For SAMM, the F_1 score is 0.3846. The overall score on the two datasets is 0.3514. The leaderboard of ME

Table 2: The effect of using denoising method. (Evaluate on CAS(ME)²).

Low-pass filter	EMD	F1 Score	Precision Score	Recall Score
×	×	0.3425	0.2841	0.4310
×	✓	0.3438	0.2860	0.4310
✓	×	0.3976	0.3443	0.4704
✓	✓	0.4000	0.3479	0.4704

and MaE spotting task can be seen in Table 1. Our method performs best in the challenge.

5 CONCLUSION

ME and MaE spotting task is often inseparable from the extraction of optical flow features. However, a persistent challenge of such methods is that the optical flow features of the whole face contain both facial expression motion information and many facial motion information unrelated to expression. We effectively remove expression-irrelevant information from the optical flow features by face cropping and alignment, optical flow extraction from the regions of interest, denoising, and sliding window method, thus making the ME and MaE spotting method based on optical flow features more accurate.

We conduct experiments on the the unseen testing set (MEGC2022-TestSet). For CAS(ME)³, the F_1 score of the proposed method is 0.3333. For SAMM, the F_1 score is 0.3846. The overall F_1 score is 0.3514 and we achieve the first place in the MEGC2022 Challenge. The results prove the effectiveness of our proposed method.

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